

Subhrajyoty Roy

roy.s@wustl.edu

EcoSta 2026

Aug 8, 2026

Robust determination of the number of factors via divergence information criteria

in collaboration with **Abhik Ghosh** *and* **Ayanendranath Basu**

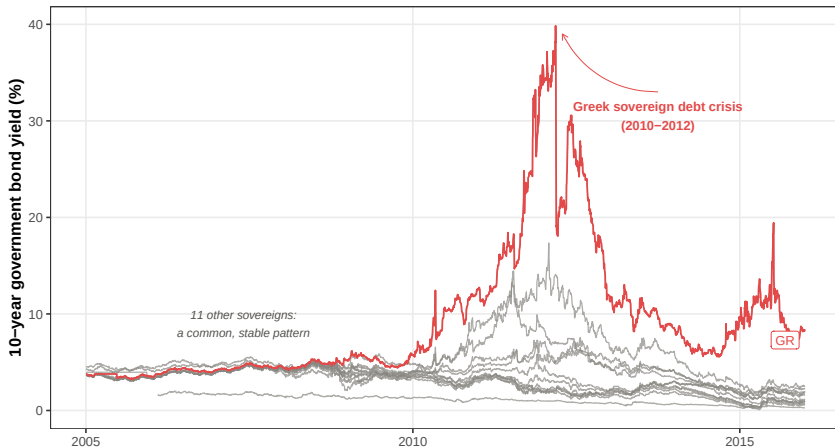
Motivation

- It is well known that the yields of sovereign bonds across countries are highly correlated.
- There are typically a few macroeconomic (global) factors that drive the yields of sovereign bonds across countries.
- Let's look at some recent data (on daily yields of 10 year bonds) to see if this is indeed the case!

Cross-Country Sovereign Bond Yields

Cross-country bond yields mostly move together – until they don't

10Y sovereign yields, 12 countries, 2005–2015



- › The above data can be represented as a matrix.
- › Rows represent time points, and columns represent countries.
- › The underlying factors provide a low-rank signal to the matrix data.

$$\mathbf{X}_{n \times p} = \mathbf{L} + \mathbf{E},$$

where $\mathbf{L}$ is a low-rank matrix of rank r , and $\mathbf{E}$ is a noise matrix with

$$E_{ij} \sim_{\text{IID}} (1 - \epsilon)f + \epsilon k$$

where f is the modelling density, and k is the contamination density.

- Recommendation systems are widely used in e-commerce platforms to recommend products to users.
- Collected data includes (user, item) pairs and corresponding ratings.
- Underlying factors reveal the preferences of users and the characteristics of items.

Recommender Systems: Shilling Attacks

- Recommendation systems are widely used in e-commerce platforms to recommend products to users.
- Collected data includes (user, item) pairs and corresponding ratings.
- Underlying factors reveal the preferences of users and the characteristics of items.

CNBC MARKETS BUSINESS INVESTING TECH POLITICS VIDEO INVESTING CLUB PRO LIVESTREAM

TECH

Amazon flooded with thousands of fake reviews, report claims

PUBLISHED TUE, APR 16 2019-5:49 AM EDT

Chloe Taylor
@CNBCDETAILED191

SHARE    

RESEARCH-ARTICLE |  FREE ACCESS | 

LLM Agent-based Shilling Attack on Recommender Systems

Authors:  Shengkang Gu,  Jiahao Liu,  Dongsheng Li,  Guangping Zhang,  Mingzhe Han,  Hansu Gu,  Peng Zhang,  Ning Gu,  Li Shang,  Tun Lu, [Authors Info & Claims](#)

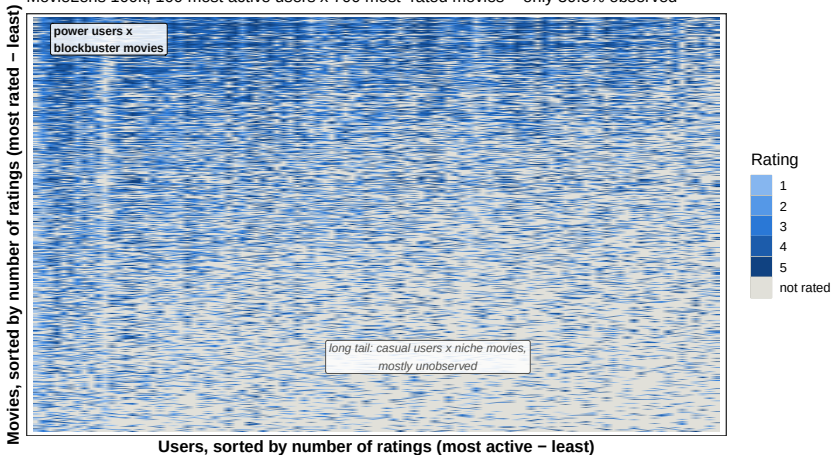
WSDM '26: Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining
Pages 163 - 173 • <https://doi.org/10.1145/3773966.3777981>

Published: 21 February 2026 [Publication History](#)  Check for updates

Recommender Systems: MovieLens Data

A recommender system's ratings matrix is mostly missing

MovieLens 100k, 100 most active users x 700 most-rated movies – only 39.5% observed



The $\mathbf{L} + \mathbf{E}$ decomposition arises across many different fields.

- 1 *Background Modelling*: In video surveillance data, background is like low-rank slowly varying signal, and the noise (or sparse components) are like foregrounds [Roy et al., 2024].
- 2 *Gene Expression*: Gene expression matrices are typically partially completed, and outliers exist due to lack of precise measurements. The low-rank component provides underlying biological factors [Linderman et al., 2022].

Core Challenges

Robust Singular Value Decomposition

- The classical SVD is not robust to outliers.
- We need a robust SVD method to estimate the low-rank component $\mathbf{L}$.

Assume that the rank r is known.

- The classical SVD is not robust to outliers.
- We need a robust SVD method to estimate the low-rank component L .

Assume that the rank r is known.

Then,

$$X_{ij} = \sum_{k=1}^r U_{ik} D_{kk} V_{jk} + E_{ij} = \sum_{k=1}^r a_{ik} b_{jk} + E_{ij},$$

Robust Singular Value Decomposition

- The classical SVD is not robust to outliers.
- We need a robust SVD method to estimate the low-rank component L .

Assume that the rank r is known.

Then,

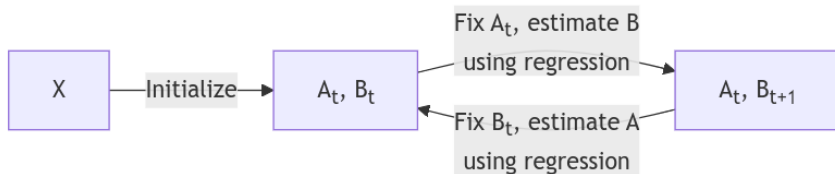
$$X_{ij} = \sum_{k=1}^r U_{ik} D_{kk} V_{jk} + E_{ij} = \sum_{k=1}^r a_{ik} b_{jk} + E_{ij},$$

Fix j , and get

$$(X_{\cdot j})_{n \times 1} = \mathbf{A}_{n \times r} (B_{j \cdot})_{r \times 1} + E_{\cdot j}$$

Assuming $\mathbf{A}$ is known, we can estimate each $(B_{j \cdot})$ by solving a **robust** regression problem.

Robust SVD: The Iterative Regression



- 1 Assume that rank r is known.
- 2 Initialize $A_0 = A$ and $B_0 = B$ using classical SVD (or random orthogonal matrices, scaled by the singular values).
- 3 Given A_t , estimate B_{t+1} by solving p robust multiple linear regression problems.
- 4 Given B_{t+1} , estimate A_{t+1} by solving n robust multiple linear regression problems.
- 5 Repeat until convergence.

Naive Implementation Since r is also the number of true non-zero regression coefficients, we can use regression variable selection methods (e.g., LASSO, SCAD, Knockoffs, etc.)

Naive Implementation Since r is also the number of true non-zero regression coefficients, we can use regression variable selection methods (e.g., LASSO, SCAD, Knockoffs, etc.)

Problem: Selected variables may be different for different choices of $j = 1, \dots, p$ – there is no single, shared r .

Naive Implementation Since r is also the number of true non-zero regression coefficients, we can use regression variable selection methods (e.g., LASSO, SCAD, Knockoffs, etc.)

Problem: Selected variables may be different for different choices of $j = 1, \dots, p$ – there is no single, shared r .

Two broad strategies dominate the literature:

- 1 **Information criteria** – penalize objective function $d(\cdot)$ analytically, in closed form.
- 2 **Computational / resampling approaches** – remove part of the data, refit, and measure prediction error on the held-out part.

$$C(r) = d(\mathbf{X}, \hat{\mathbf{L}}_r) + ra(\hat{\sigma})g(n, p).$$

The rank is estimated as $\hat{r} = \arg \min_r C(r)$, and $\hat{\mathbf{L}}_r$ from the rank r approximation from SVD, $a(\cdot)$ is an appropriate function, and $g(\cdot)$ is the penalty.

$$C(r) = d(\mathbf{X}, \hat{\mathbf{L}}_r) + ra(\hat{\sigma})g(n, p).$$

The rank is estimated as $\hat{r} = \arg \min_r C(r)$, and $\hat{\mathbf{L}}_r$ from the rank r approximation from SVD, $a(\cdot)$ is an appropriate function, and $g(\cdot)$ is the penalty.

- ▶ Examples: AIC [Akaike, 1973], BIC [Schwarz, 1978], and the panel-data criteria PC1–PC3, IC1–IC3 of [Bai and Ng, 2002].
- ▶ A related family: hard/soft **thresholding** of singular values [Shabalin and Nobel, 2013, Gavish and Donoho, 2014, Choi et al., 2017, Xu et al., 2021].

Approach 1: Information Criteria

$$C(r) = d(\mathbf{X}, \hat{\mathbf{L}}_r) + ra(\hat{\sigma})g(n, p).$$

The rank is estimated as $\hat{r} = \arg \min_r C(r)$, and $\hat{\mathbf{L}}_r$ from the rank r approximation from SVD, $a(\cdot)$ is an appropriate function, and $g(\cdot)$ is the penalty.

➤ Examples: AIC [Akaike, 1973], BIC [Schwarz, 1978], and the panel-data criteria PC1–PC3, IC1–IC3 of [Bai and Ng, 2002].

➤ A related family: hard/soft **thresholding** of singular values [Shabalin and Nobel, 2013, Gavish and Donoho, 2014, Choi et al., 2017, Xu et al., 2021].

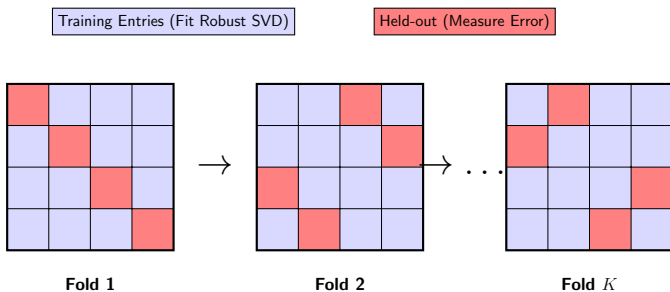
✗ **Fast, closed-form** – but classical criteria is built on $\|\mathbf{X} - \hat{\mathbf{L}}_r\|_F^2$: **a single outlier can dominate this term** and derail the whole criterion, *even if $\hat{\mathbf{L}}_r$ itself comes from a robust SVD.*

Approach 2: Computational Approach

- **Cross-validation / resampling:** hold out a subset of entries (or rows/columns), refit a rank- r approximation on the rest, and measure prediction error on the held-out part – repeat over folds and choose r minimizing the prediction error.

Approach 2: Computational Approach

- Wold-style (or speckled) [Wold, 1978]. The removal of entries happen at each cell / entry level.

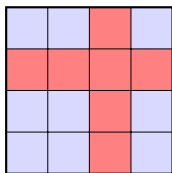


Approach 2: Computational Approach

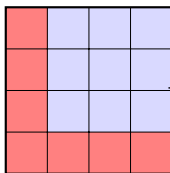
- Eastment–Krzanowski [Eastment and Krzanowski, 1982], Gabriel-style [Gabriel, 2002], and bi-cross-validation [Owen and Perry, 2009]. The removal of entries happen at each row / column / block levels.

Training Entries (Fit Robust SVD)

Held-out Rows/Cols (Measure Error)



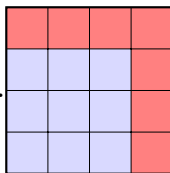
Fold 1



Fold 2



...



Fold K

- Using classical SVD leads to non-robust selection of rank.
- Can be robustified by swapping in a robust SVD at every fold [Arciniegas-Alarcón et al., 2022] ...
- ✗ ... but this is computationally prohibitive:
 - Gabriel-style CV on a modest 50×40 matrix needs ≈ 2000 SVDs.
 - At ≈ 1 second per robust SVD, rank estimation alone takes ≈ 33 minutes.

- ▶ Using classical SVD leads to non-robust selection of rank.
- ▶ Can be robustified by swapping in a robust SVD at every fold [Arciniegas-Alarcón et al., 2022] ...
- × ... but this is computationally prohibitive:
 - ▶▶ Gabriel-style CV on a modest 50×40 matrix needs ≈ 2000 SVDs.
 - ▶▶ At ≈ 1 second per robust SVD, rank estimation alone takes ≈ 33 minutes.
- ▶ One idea is to perform one full-rank (i.e., upto $\min(n, p)$) $\widehat{\mathbf{X}}$ from a single robust SVD of $\mathbf{X}$, which replaces outliers with reasonable predictions. Then apply cross-validation with classical SVD on $\widehat{\mathbf{X}}$, as it is free of outliers.

- **The “Elbow” method** [Thorndike, 1953]: eyeball the scree plot of singular values for a visible drop. Simple, but subjective, and ill-defined under contamination.
- **Bayesian model averaging** over the rank and the SVD components [Hoff, 2007].
- **Spectral regularization** / nuclear-norm penalization with soft-imputation [Mazumder et al., 2010].

- **The “Elbow” method** [Thorndike, 1953]: eyeball the scree plot of singular values for a visible drop. Simple, but subjective, and ill-defined under contamination.
- **Bayesian model averaging** over the rank and the SVD components [Hoff, 2007].
- **Spectral regularization** / nuclear-norm penalization with soft-imputation [Mazumder et al., 2010].
- ✗ None of these were designed with robustness to outliers in mind.

Divergence Information Criterion for Matrix Rank

Definition

[Basu et al., 1998] defined the *density power divergence* between two probability densities f and g as

$$d_{\alpha}(f, g) = \int f^{1+\alpha} - \left(1 + \frac{1}{\alpha}\right) \int f^{\alpha} g + \frac{1}{\alpha} \int g^{\alpha},$$

where $\alpha > 0$ is the robustness parameter.

Definition

[Basu et al., 1998] defined the *density power divergence* between two probability densities f and g as

$$d_{\alpha}(f, g) = \int f^{1+\alpha} - \left(1 + \frac{1}{\alpha}\right) \int f^{\alpha} g + \frac{1}{\alpha} \int g^{\alpha},$$

where $\alpha > 0$ is the robustness parameter.

In the Robust SVD context, let f be the model density, $\theta = (U, D, V, \sigma^2)$ be parameters from SVD. Then,

$$H_{\alpha}(\theta) = \frac{1}{\sigma^{\alpha}} \left[\int f^{1+\alpha} - \left(1 + \frac{1}{\alpha}\right) \frac{1}{np} \sum_{i=1}^n \sum_{j=1}^p f^{\alpha} \left(\frac{X_{ij} - (UDV^{\top})_{ij}}{\sigma} \right) \right].$$

DIC is simply unbiased estimate of $Q_\alpha(\hat{\theta})$, where

$$Q_\alpha(\theta) = \mathbb{E}_{\mathbf{X}}(H_\alpha(\theta)), \text{ for all } \theta \in \Theta$$

DIC is simply unbiased estimate of $Q_\alpha(\hat{\theta})$, where

$$Q_\alpha(\theta) = \mathbb{E}_{\mathbf{X}}(H_\alpha(\theta)), \text{ for all } \theta \in \Theta$$

$$Q_\alpha(\hat{U}, \hat{D}, \hat{V}, \hat{\sigma}^2) = \mathbb{E} \left(Q_\alpha(\hat{U}, \hat{D}, \mathbf{V}, \hat{\sigma}^2) \mid \mathbf{V} = \hat{\mathbf{V}} \right)$$

DIC is simply unbiased estimate of $Q_\alpha(\hat{\theta})$, where

$$Q_\alpha(\theta) = \mathbb{E}_{\mathbf{X}}(H_\alpha(\theta)), \text{ for all } \theta \in \Theta$$

$$\begin{aligned} Q_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}, \hat{\mathbf{V}}, \hat{\sigma}^2) &= \mathbb{E} \left(Q_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}, \mathbf{V}, \hat{\sigma}^2) \mid \mathbf{V} = \hat{\mathbf{V}} \right) \\ &= \mathbb{E} \left(\mathbb{E} \left(H_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}) + \frac{1}{n} \hat{\sigma}^{-\alpha} \text{trace}(\mathbf{V}^\top \mathbf{V}) \frac{C_{2\alpha}}{C_\alpha} \mid \mathbf{V} = \hat{\mathbf{V}} \right) \right) \end{aligned}$$

DIC is simply unbiased estimate of $Q_\alpha(\hat{\theta})$, where

$$Q_\alpha(\theta) = \mathbb{E}_{\mathbf{X}}(H_\alpha(\theta)), \text{ for all } \theta \in \Theta$$

$$\begin{aligned} Q_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}, \hat{\mathbf{V}}, \hat{\sigma}^2) &= \mathbb{E} \left(Q_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}, \mathbf{V}, \hat{\sigma}^2) \mid \mathbf{V} = \hat{\mathbf{V}} \right) \\ &= \mathbb{E} \left(\mathbb{E} \left(H_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}) + \frac{1}{n} \hat{\sigma}^{-\alpha} \text{trace}(\mathbf{V}^\top \mathbf{V}) \frac{C_{2\alpha}}{C_\alpha} \mid \mathbf{V} = \hat{\mathbf{V}} \right) \right) \\ &= \mathbb{E} \left(H_\alpha(\hat{\theta}) + \frac{r}{n} \hat{\sigma}^{-\alpha} \frac{C_{2\alpha}}{C_\alpha} \right) \end{aligned}$$

DIC is simply unbiased estimate of $Q_\alpha(\hat{\theta})$, where

$$Q_\alpha(\theta) = \mathbb{E}_{\mathbf{X}}(H_\alpha(\theta)), \text{ for all } \theta \in \Theta$$

$$\begin{aligned} Q_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}, \hat{\mathbf{V}}, \hat{\sigma}^2) &= \mathbb{E} \left(Q_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}, \mathbf{V}, \hat{\sigma}^2) \mid \mathbf{V} = \hat{\mathbf{V}} \right) \\ &= \mathbb{E} \left(\mathbb{E} \left(H_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}) + \frac{1}{n} \hat{\sigma}^{-\alpha} \text{trace}(\mathbf{V}^\top \mathbf{V}) \frac{C_{2\alpha}}{C_\alpha} \mid \mathbf{V} = \hat{\mathbf{V}} \right) \right) \\ &= \mathbb{E} \left(H_\alpha(\hat{\theta}) + \frac{r}{n} \hat{\sigma}^{-\alpha} \frac{C_{2\alpha}}{C_\alpha} \right) \\ &= \mathbb{E} \left(H_\alpha(\hat{\theta}) + \frac{r}{p} \hat{\sigma}^{-\alpha} \frac{C_{2\alpha}}{C_\alpha} \right), \text{ by symmetry} \end{aligned}$$

DIC is simply unbiased estimate of $Q_\alpha(\hat{\theta})$, where

$$Q_\alpha(\theta) = \mathbb{E}_{\mathbf{X}}(H_\alpha(\theta)), \text{ for all } \theta \in \Theta$$

$$\begin{aligned} Q_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}, \hat{\mathbf{V}}, \hat{\sigma}^2) &= \mathbb{E} \left(Q_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}, \mathbf{V}, \hat{\sigma}^2) \mid \mathbf{V} = \hat{\mathbf{V}} \right) \\ &= \mathbb{E} \left(\mathbb{E} \left(H_\alpha(\hat{\mathbf{U}}, \hat{\mathbf{D}}) + \frac{1}{n} \hat{\sigma}^{-\alpha} \text{trace}(\mathbf{V}^\top \mathbf{V}) \frac{C_{2\alpha}}{C_\alpha} \mid \mathbf{V} = \hat{\mathbf{V}} \right) \right) \\ &= \mathbb{E} \left(H_\alpha(\hat{\theta}) + \frac{r}{n} \hat{\sigma}^{-\alpha} \frac{C_{2\alpha}}{C_\alpha} \right) \\ &= \mathbb{E} \left(H_\alpha(\hat{\theta}) + \frac{r}{p} \hat{\sigma}^{-\alpha} \frac{C_{2\alpha}}{C_\alpha} \right), \text{ by symmetry} \\ &= \mathbb{E} \left(\underbrace{H_\alpha(\hat{\theta}) + r \frac{(1/n + 1/p)}{2} \hat{\sigma}^{-\alpha} \frac{C_{2\alpha}}{C_\alpha}}_{\text{DICMR}_{\alpha,1}(r)} \right) \end{aligned}$$

- 1 The model density f is symmetric about 0, decreasing in $|x|$ and is differentiable in $|x|$.
- 2 $\sup_{x \in \mathbb{R}} \max \{|f^\alpha(x) \nabla \log(f(x))|, |x f^\alpha(x) \nabla \log(f(x))|\} < \infty$.
- 3 True noise variance $\sigma^2 \asymp (np)^{-1/2}$.
- 4 Estimated noise variance $\hat{\sigma}^2$ lies between $[\sigma_{\min}^2, \sigma_{\max}^2]$ with $\sigma_{\min}^2 \asymp (np)^{-1/2}$ and $\sigma_{\max}^2 \asymp 1$.

- 1 The model density f is symmetric about 0, decreasing in $|x|$ and is differentiable in $|x|$.
- 2 $\sup_{x \in \mathbb{R}} \max \{|f^\alpha(x) \nabla \log(f(x))|, |x f^\alpha(x) \nabla \log(f(x))|\} < \infty$.
- 3 True noise variance $\sigma^2 \asymp (np)^{-1/2}$.
- 4 Estimated noise variance $\hat{\sigma}^2$ lies between $[\sigma_{\min}^2, \sigma_{\max}^2]$ with $\sigma_{\min}^2 \asymp (np)^{-1/2}$ and $\sigma_{\max}^2 \asymp 1$.

Standard f ensures (1) and (2) holds, e.g., Gaussian, t , logistic, etc.

Assumption 3 is **necessary**, [Shabalin and Nobel, 2013] showed with any larger error variance, the singular values (or their functions) cannot be consistently estimated.

Theorem (Simplified)

Consider a criterion of the form

$$C_{\alpha}(r) = H_{\alpha}(\hat{\theta}) + r\gamma_{n,p}\hat{\sigma}^{-\alpha},$$

where $\gamma_{n,p} > 0$ is a sequence of positive constants. Suppose $r < r^$, the true rank, and $\sigma_{r,\alpha}^* \geq \sigma(1 + O(1))$, $\gamma_{n,p} = o(1)$. Then,*

$$\mathbb{P}(C_{\alpha}(r) \leq C_{\alpha}(r^*)) \leq C_1 \exp(-npC_2 + C_3(n + p + 1)r^* \log(np)),$$

where C_1, C_2 and C_3 are some generic constants independent of n and p .

Theorem (Simplified)

Consider a criterion of the form

$$C_\alpha(r) = H_\alpha(\hat{\theta}) + r\gamma_{n,p}\hat{\sigma}^{-\alpha},$$

where $\gamma_{n,p} > 0$ is a sequence of positive constants. Pick any $r \in (r^, r_{\max}]$, where r^* is the true rank. Then,*

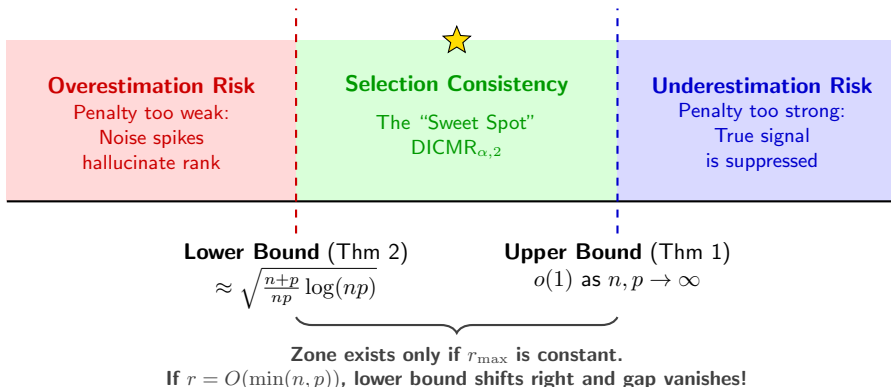
$$\mathbb{P}(C_\alpha(r) \leq C_\alpha(r^*)) \leq \max\{B(r_{\max}), B(r^*)\},$$

where

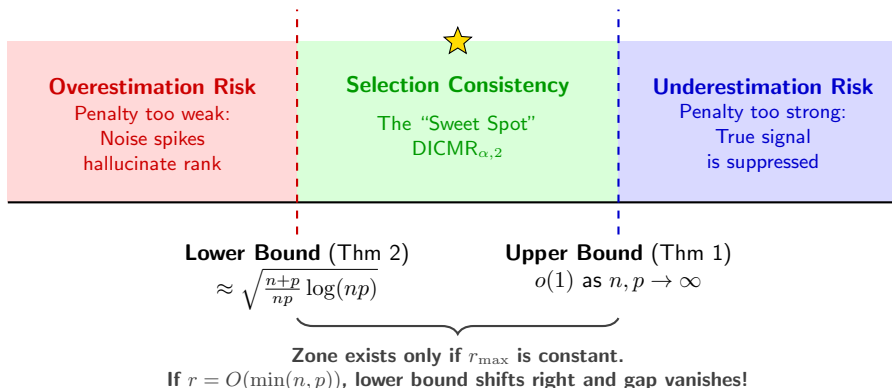
$$B(r) = C_5 \exp(-npC_6\gamma_{n,p}^2 + C_7((n+p+1)r+1)\log(np))$$

for some appropriate positive constants C_5, C_6, C_7 free of n, p and r , depending on the true rank r^ .*

The Second variant of DICMR



The Second variant of DICMR



$$\text{DICMR}_{\alpha,2}(r) := H_{r,\alpha}(\hat{\boldsymbol{\theta}}_{r,\alpha}) + \frac{r\sqrt{(n+p)\log(np)}}{2\sqrt{np}} \hat{\sigma}_{r,\alpha}^{-\alpha} C_{2\alpha}^f / C_{\alpha}^f$$

Theorem

Consider a criterion of the form

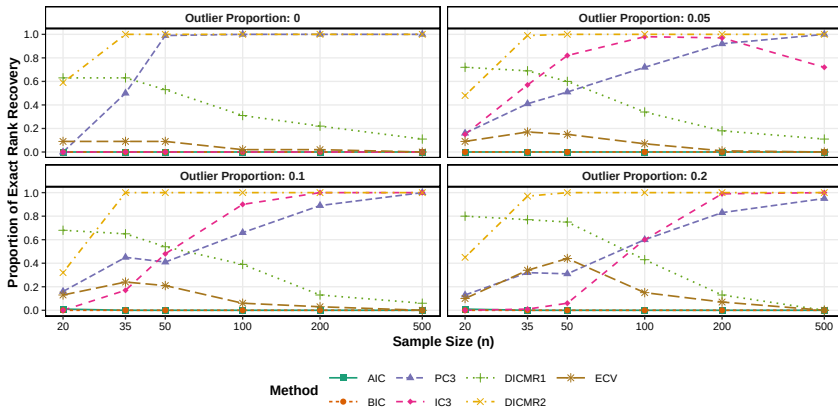
$$C_{\alpha}(r) = H_{\alpha}^{(r)}(\hat{\theta}) + r\gamma_{n,p}\hat{\sigma}^{-\alpha},$$

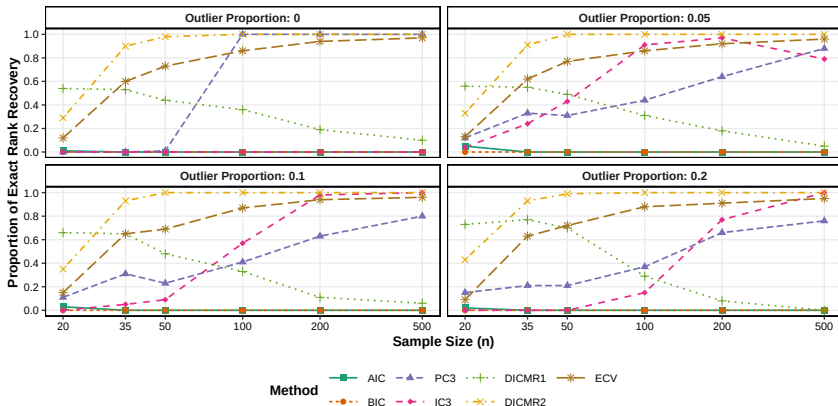
where $\gamma_{n,p} > 0$ is a sequence of positive constants. The model selection criterion $C_{\alpha}(r)$ is first-order B-robust for any $\alpha > 0$, provided that the corresponding rSVDdpd estimator (i.e., the iterative method to estimate $\hat{\theta}$ by minimizing DPD) is B-robust, and $\gamma_{n,p} = O(1)$.

Experimental Results

- Random orthogonal matrices as left (and right) singular vectors.
- Two settings: (i) First r singular values are 1, rest are 0. (ii) The singular values are $2, 1 + (r - 2)/(r - 1), \dots, 1$, and rest are 0.
- Noise entries are i.i.d. normal with mean zero and variance $\sigma_e^2 = \mathbb{E}(\|\mathbf{L}\|_F^2)/(np \times \text{SNR})$.
- SNR levels are 2, 4 and 10. Randomly chosen 5%, 10% or 20% entries are contaminated with a value twice of the maximum.

Simulation Results





- **Generalized Criterion:** Asymptotic consistency holds for a family of criteria scaled by a constant $c > 0$:

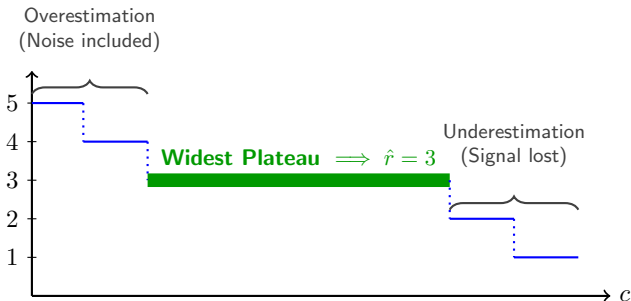
$$\tilde{C}_\alpha(r) = H_{r,\alpha}(\hat{\boldsymbol{\theta}}_{r,\alpha}) + c \cdot r \hat{\sigma}_{r,\alpha}^{-\alpha} \frac{\sqrt{(n+p) \log(np)}}{\sqrt{np}}$$

- **The Challenge:** For finite n and p , the penalty requires careful tuning.
 - c too small $\implies$ Penalty too weak (Overestimation)
 - c too large $\implies$ Penalty too aggressive (Underestimation)

Finite-Sample Tuning: The Plateau Heuristic

The Plateau Heuristic: Evaluate optimal rank $r(c)$ over a grid of c . The true rank is highly stable against perturbations, corresponding to the **widest flat region** [Hallin and Liška, 2007].

Selected Rank $r(c)$

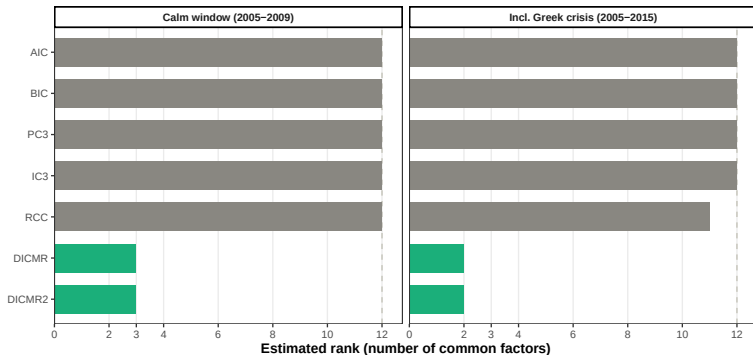


Bond Yield Data: Classical Criteria Find Nothing

- Panel of 10Y sovereign yields, 12 countries.
- Split into a calm window (2005–2009) and a window spanning the 2010–2012 Greek crisis (2005–2015).

Classical criteria find no structure at all; DICMR recovers a compact factor model

10Y sovereign yield panel (12 countries) -- classical AIC/BIC/PC/IC vs. DICMR

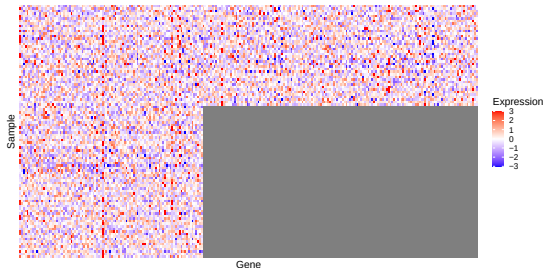


Dashed line: ceiling rank $p = 12$, i.e. no compression at all

- Every classical criterion picks $\hat{r} = p = 12$ in **both** windows: with only 12 countries but thousands of trading days ($n \gg p$), their penalty is too weak to ever compress – they treat every country as its own factor.
- DICMR / DICMR₂ instead recover a compact structure in both regimes: rank 3 in the calm window, rank 2 once the Greek crisis is included – consistent with the textbook level/slope/curvature factor structure of yield curves. (e.g., *Inflation, Central Bank Policy, Economic Activity, etc.*)

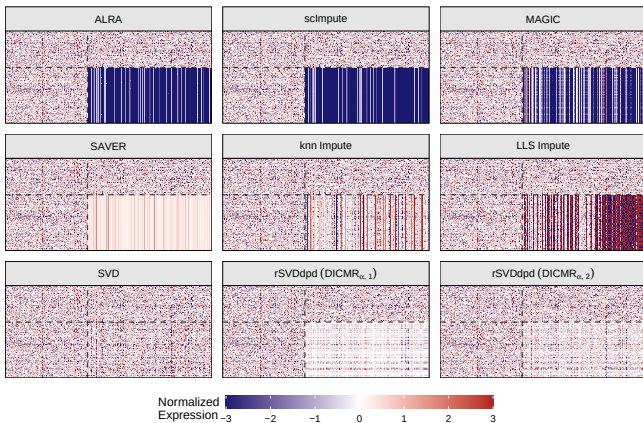
Gene Expression: Why It Matters

- RNA-sequencing data: rows are patient samples, columns are genes – a handful of shared biological factors should drive a low-rank signal.
- Sequencing is costly, and combining multiple studies creates **block-missing** patterns (some patients profiled on all genes, some genes profiled on all patients) – a genuine matrix completion problem, not just random missingness.
- TCGA PAN-Cancer RNA-seq dataset [Fiorini, 2016]: 801 patient samples $\times$ 20,531 genes, spanning 5 cancer types, log2-normalized.



Gene Expression: Completions Compared

- Train on a random 100×2000 block $\mathbf{Z}_{11}$; predict the held-out block $\mathbf{X}_{22}$ (below the dashed line) via $\hat{\mathbf{X}}_{22} = \mathbf{X}_{21}[\hat{\mathbf{L}}_{r,\alpha}]_{11}^+ \mathbf{X}_{12}$.
- Several specialized methods collapse the held-out block to a near-constant value or show streaky artifacts; classical SVD and rSVDdpd stay coherent with the observed pattern.



Gene Expression: Accuracy Comparison

Method	Relative RMSE
llsImpute [Kim et al., 2005]	1.2418
sclImpute [Li and Li, 2018]	0.7497
ALRA [Linderman et al., 2022]	0.7468
MAGIC [van Dijk et al., 2018]	0.0881
knnImpute [Troyanskaya et al., 2001]	0.0365
SAVER [Huang et al., 2018]	0.0256
SVD	0.0215
rSVDdpd ($\text{DICMR}_{\alpha,1}$); $\hat{r} = 2$ (ours)	0.0205
rSVDdpd ($\text{DICMR}_{\alpha,2}$); $\hat{r} = 7$ (ours)	0.0186

Gene Expression: Accuracy Comparison

Method	Relative RMSE
llsImpute [Kim et al., 2005]	1.2418
sclmpute [Li and Li, 2018]	0.7497
ALRA [Linderman et al., 2022]	0.7468
MAGIC [van Dijk et al., 2018]	0.0881
knnImpute [Troyanskaya et al., 2001]	0.0365
SAVER [Huang et al., 2018]	0.0256
SVD	0.0215
rSVDdpd (DICMR $_{\alpha,1}$); $\hat{r} = 2$ (ours)	0.0205
rSVDdpd (DICMR $_{\alpha,2}$); $\hat{r} = 7$ (ours)	0.0186

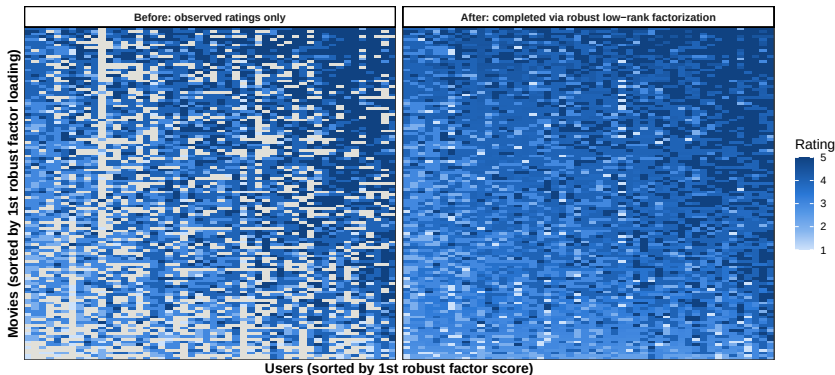
- Relative RMSE = $\|\mathbf{X}_{22} - \hat{\mathbf{X}}_{22}\|_F^2 / \|\mathbf{X}_{22}\|_F^2$, on the original (un-normalized) scale.
- ✓ rSVDdpd + DICMR $_{\alpha,2}$ gives the lowest error; closest contenders are SAVER, classical SVD, and knnImpute.

Recommender Systems: Filling in the Blanks

- ▶ MovieLens 100k, restricted to 50 active users $\times$ 120 popular movies (75% observed).
- ▶ Rows/columns sorted by their loading on the first robust factor – the same sort reveals a smooth gradient in *both* panels.

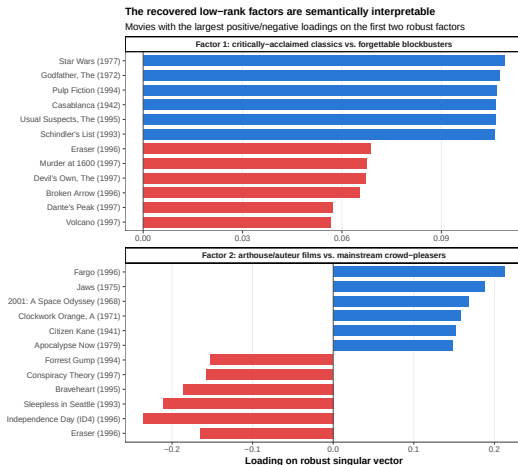
Filling in the missing ratings under a low-rank assumption

MovieLens 100k, 50 users $\times$ 120 movies -- 75.1% observed, completed at rank 2 (rSVDdpd)



Recommender Systems: Are the Factors Meaningful?

- A numerically low rank is only useful if the recovered factors correspond to something real – so look at which movies load most heavily on each robust singular vector.



- Factor 1 separates critically-acclaimed classics (*Godfather*, *Pulp Fiction*, *Casablanca*) from forgettable mid-90s blockbusters (*Volcano*, *Dante's Peak*).

Recommender Systems: Interpreting the Factors

- Factor 1 separates critically-acclaimed classics (*Godfather*, *Pulp Fiction*, *Casablanca*) from forgettable mid-90s blockbusters (*Volcano*, *Dante's Peak*).
- Factor 2 separates arthouse/auteur films (*Fargo*, *2001: A Space Odyssey*, *Citizen Kane*) from mainstream crowd-pleasers (*Independence Day*, *Forrest Gump*).

Recommender Systems: Interpreting the Factors

- Factor 1 separates critically-acclaimed classics (*Godfather*, *Pulp Fiction*, *Casablanca*) from forgettable mid-90s blockbusters (*Volcano*, *Dante's Peak*).
- Factor 2 separates arthouse/auteur films (*Fargo*, *2001: A Space Odyssey*, *Citizen Kane*) from mainstream crowd-pleasers (*Independence Day*, *Forrest Gump*).
- ✓ The robust low-rank structure recovers the “taste dimensions” typically found in the recommender-systems literature.

- Rank estimation for noisy matrices has two standard tools – information criteria and cross-validation – and **neither is a good fit under contamination**: classical criteria break even with a robust SVD plugged in, and robustified cross-validation is computationally prohibitive.

- Rank estimation for noisy matrices has two standard tools – information criteria and cross-validation – and **neither is a good fit under contamination**: classical criteria break even with a robust SVD plugged in, and robustified cross-validation is computationally prohibitive.
- **DICMR**: a closed-form penalized criterion built on the density power divergence, inheriting robustness directly from the rSVDdpd estimator.

- Rank estimation for noisy matrices has two standard tools – information criteria and cross-validation – and **neither is a good fit under contamination**: classical criteria break even with a robust SVD plugged in, and robustified cross-validation is computationally prohibitive.
- **DICMR**: a closed-form penalized criterion built on the density power divergence, inheriting robustness directly from the $rSVDdpd$ estimator.
- Non-asymptotic bounds pin down exactly when DICMR avoids both over- and underestimation, and it is first-order B-robust by construction.

- Rank estimation for noisy matrices has two standard tools – information criteria and cross-validation – and **neither is a good fit under contamination**: classical criteria break even with a robust SVD plugged in, and robustified cross-validation is computationally prohibitive.
- **DICMR**: a closed-form penalized criterion built on the density power divergence, inheriting robustness directly from the $rSVD_{dpd}$ estimator.
- Non-asymptotic bounds pin down exactly when DICMR avoids both over- and underestimation, and it is first-order B-robust by construction.
- Across three very different real datasets – cross-country bond yields, TCGA gene expression, and MovieLens ratings – DICMR recovers compact, interpretable, and more accurate low-rank structure than classical or resampling alternatives, at a fraction of the computational cost.



Hallin, Marc, and Roman Liška. "Determining the number of factors in the general dynamic factor model." *Journal of the American Statistical Association* 102, no. 478 (2007): 603-617.



Basu, Ayanendranath, Ian R. Harris, Nils L. Hjort, and Michael C. Jones. "Robust and efficient estimation by minimising a density power divergence." *Biometrika* 85, no. 3 (1998): 549-559.



Linderman, George C., Jun Zhao, Manolis Roulis, Piotr Bielecki, Richard A. Flavell, Boaz Nadler, and Yuval Kluger. "Zero-preserving imputation of single-cell RNA-seq data." *Nature communications* 13, no. 1 (2022): 192.



Fiorini, Samuele. "Gene Expression Cancer RNA-Seq." *UCI Machine Learning Repository* (2016).



Kim, Hyunsoo, Gene H. Golub, and Haesun Park. "Missing value estimation for DNA microarray gene expression data: local least squares imputation." *Bioinformatics* 21.2 (2005): 187-198.



Li, Wei Vivian, and Jingyi Jessica Li. "An accurate and robust imputation method scImpute for single-cell RNA-seq data." *Nature communications* 9.1 (2018): 997.



van Dijk, David, et al. "Recovering gene interactions from single-cell data using data diffusion." *Cell* 174.3 (2018): 716-729.



Huang, Mo, et al. "SAVER: gene expression recovery for single-cell RNA sequencing." *Nature methods* 15.7 (2018): 539-542.



Troyanskaya, Olga, et al. "Missing value estimation methods for DNA microarrays." *Bioinformatics* 17.6 (2001): 520-525.



Roy, Subhrajyoty, Abhik Ghosh, and Ayanendranath Basu. "Robust singular value decomposition with application to video surveillance background modelling." *Statistics and Computing* 34.5 (2024): 178.



Akaike, Hirotugu. "Information theory and an extension of the maximum likelihood principle." *Selected Papers of Hirotugu Akaike*. Springer, New York, NY, 1973. 199-213.



Schwarz, Gideon. "Estimating the dimension of a model." *The Annals of Statistics* 6.2 (1978): 461-464.



Bai, Jushan, and Serena Ng. "Determining the number of factors in approximate factor models." *Econometrica* 70.1 (2002): 191-221.



Shabalin, Andrey A., and Andrew B. Nobel. "Reconstruction of a low-rank matrix in the presence of Gaussian noise." *Journal of Multivariate Analysis* 118 (2013): 67-76.



Gavish, Matan, and David L. Donoho. "The optimal hard threshold for singular values is $4/\sqrt{3}$." *IEEE Transactions on Information Theory* 60.8 (2014): 5040-5053.



Choi, Yunjin, Jonathan Taylor, and Robert Tibshirani. "Selecting the number of principal components: Estimation of the true rank of a noisy matrix." *The Annals of Statistics* 45.6 (2017): 2590-2617.



Xu, Zhengqin, Rui He, Shoulie Xie, and Shiqian Wu. "Adaptive rank estimate in robust principal component analysis." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2021. 6577-6586.



Wold, Svante. "Cross-validatory estimation of the number of components in factor and principal components models." *Technometrics* 20.4 (1978): 397-405.



Eastment, H. T., and W. J. Krzanowski. "Cross-validatory choice of the number of components from a principal component analysis." *Technometrics* 24.1 (1982): 73-77.



Gabriel, K. Ruben. "Le biplot – outil d'exploration de données multidimensionnelles." *Journal de la Société française de statistique* 143.3-4 (2002): 5-55.



Owen, Art B., and Patrick O. Perry. "Bi-cross-validation of the SVD and the nonnegative matrix factorization." *The Annals of Applied Statistics* 3.2 (2009): 564-594.



Arciniegas-Alarcón, Sergio, Marisol García-Peña, and Wojtek J. Krzanowski. "Cross-validation for lower rank matrices containing outliers." *Applied System Innovation* 5.4 (2022): 69.



Thorndike, Robert L. "Who belongs in the family?" *Psychometrika* 18.4 (1953): 267-276.



Hoff, Peter D. "Model averaging and dimension selection for the singular value decomposition." *Journal of the American Statistical Association* 102.478 (2007): 674-685.



Mazumder, Rahul, Trevor Hastie, and Robert Tibshirani. "Spectral regularization algorithms for learning large incomplete matrices." *The Journal of Machine Learning Research* 11 (2010): 2287-2322.

Subhrajyoty Roy
roy.s@wustl.edu



Paper: <https://arxiv.org/abs/2510.19583>